

Course title: Research in Education

Course No. : Ed 462

Level: B.Ed. ICT

Semester: Sixth

Nature of Course: Theoretical

Credit Hours: 3

Teaching hours: 48

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## 1. Course Description

This course is designed to provide students with basic understanding and skills of educational research. It helps students understand meaning and types of educational research, select appropriate strategies for carrying out educational research and prepare research proposal and report. In addition, this course helps students distinguish between qualitative and quantitative research and select appropriate data collection and analysis strategies for both types of research. It further familiarises students with the meaning of action research and the process of carrying out action research in educational setting. The course prepares the students to carry out small-scale educational research.

## 2. General Objectives

The general objectives of the course are as follows:

- To provide students with understanding of research and educational research.
- To acquaint students with steps in educational research
- To help students understand the meaning of quantitative research and enable them to carry out quantitative research.
- To enable students understand the meaning of qualitative research and to carry out qualitative research.
- To acquaint students with the meaning of action research and enable them to carry out action research.
- To enable students to prepare research proposal and research report.

## 3. Specific objectives and contents

| Specific Objectives                                                                                                                                                                                                                               | Contents                                                                                                                                                                                                     |
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| <ul style="list-style-type: none"><li>• Discuss the meaning of research and educational research.</li><li>• Select research problems from the educational field.</li><li>• List steps in educational research and explain each of them.</li></ul> | <b>Unit I Introduction to Educational Research (8)</b><br><br>1.1 Meaning of research and educational research<br>1.2 Research issues and problems in educational field<br>1.3 Steps in educational research |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.3.1 Selecting a problem<br>1.3.2 Reviewing the literature<br>1.3.3 Designing the research<br>1.3.4 Collecting the data<br>1.3.5 Analysing the data<br>1.3.6 Interpreting the findings<br>1.3.7 Drawing conclusions                                                                                                                                                                                                                           |
| <ul style="list-style-type: none"> <li>• Provide the meaning and characteristics of quantitative research.</li> <li>• Explain the quantitative sampling methods and use them in educational research.</li> <li>• Develop the data collection tools for quantitative research.</li> <li>• Analyse the quantitative data using simple statistics.</li> <li>• Distinguish among experimental, survey and correlational research.</li> </ul> | <b>Unit II Quantitative Research (12)</b><br><br>3.1 Meaning and characteristics of quantitative research<br>3.2 Random sampling: Simple and stratified<br>3.3 Data collection tools: Questionnaire, test, scales and check list<br>3.4 Statistical analysis: frequency, percentage, mean and standard deviation<br>3.5 Types of quantitative research<br>3.5.1 Experimental research<br>3.5.2 Survey research<br>3.5.3 Correlational Research |
| <ul style="list-style-type: none"> <li>• Provide the meaning and characteristics of qualitative research.</li> <li>• Explain the types of purposive sampling appropriate in qualitative research.</li> <li>• Use the coding and thematic analysis techniques for analysing the qualitative data.</li> <li>• Distinguish between ethnography and narrative study with examples.</li> </ul>                                                | <b>Unit III. Qualitative Research (12)</b><br><br>3.1 Meaning and characteristics of qualitative research<br>3.2 Purposive sampling<br>3.3 Data collection strategies: semi-structured and unstructured interview, participant observation, focus group discussion<br>3.4 Coding and thematic analysis<br>3.5 Types of qualitative research<br>3.5.1 Ethnography<br>3.5.2 Narrative study                                                      |
| <ul style="list-style-type: none"> <li>• Explain the meaning and characteristics of action research.</li> <li>• Discuss the steps of carrying out action research.</li> <li>• Provide examples of action research from the educational field.</li> <li>• Explain the process of action research report.</li> </ul>                                                                                                                       | <b>Unit IV Action Research (6)</b><br><br>4.1 Meaning and characteristics of action research<br>4.2 Action research cycle<br>4.2.1 Planning<br>4.2.2 Acting<br>4.2.3 Observing<br>4.2.4 Reflecting<br>4.3 Preparing action research report                                                                                                                                                                                                     |

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| <ul style="list-style-type: none"> <li>• Discuss the purpose of writing research proposal.</li> <li>• List and explain the various components of research proposal.</li> <li>• List and explain the various components of research report.</li> <li>• Use citation and referencing using APA format in preparing the research report.</li> </ul> | <b>Unit V Writing research proposal and research report (10)</b> <ul style="list-style-type: none"> <li>5.1 Purpose of writing research proposal</li> <li>5.2 Components of research proposal <ul style="list-style-type: none"> <li>5.2.1 Introduction: background, research problem, objectives, delimitation</li> <li>5.2.2 Literature review</li> <li>5.2.3 Methodology: design, sampling, data collection tools and data analysis</li> </ul> </li> <li>5.3 Components of research report <ul style="list-style-type: none"> <li>5.3.1 Introduction</li> <li>5.3.2 Literature review</li> <li>5.3.3 Methodology</li> <li>5.3.4 Analysis and findings</li> <li>5.3.5 Summary, conclusion and implications</li> <li>5.3.6 References</li> <li>5.3.7 Appendices</li> </ul> </li> <li>5.4 Citation and referencing using APA style</li> </ul> |
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#### 4. Instructional Techniques

##### 4.1 General Instructional Techniques

- Introductory presentation on each topic of the unit by the teacher
- Use of lecture, question answer, discussion, brainstorming and buzz sessions for the theoretical contents.

##### 4.2 Specific Instructional Techniques

| Unit | Suggested specific instructional techniques                                                                                                                                                                                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I    | <ul style="list-style-type: none"> <li>• Students will individually prepare at least 5 research problems and share them in the class. The teacher will provide feedback on their works.</li> </ul>                                                                                                                                                                     |
| II   | <ul style="list-style-type: none"> <li>• Students will be divided in groups and each group will select a research problem appropriate for quantitative research. They will devise a set of questionnaire/rating scale/check list for collecting the data. Each group will present the tools in the class. The teacher will provide feedback on their works.</li> </ul> |
| III  | <ul style="list-style-type: none"> <li>• Students will individually conduct one qualitative interview, transcribe the interview and develop 2-3 themes using coding and thematic analysis technique. The students will share their works in the class. The teacher will provide feedback on their works.</li> </ul>                                                    |

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| IV | <ul style="list-style-type: none"> <li>Students will be divided into groups and each group will select an educational problem for action research. They will prepare a plan to carry out the action research. Finally, they will share their plans in the whole group. The teacher will provide feedback on their works.</li> </ul> |
| V  | <ul style="list-style-type: none"> <li>Students will be divided into groups and they will select a research topic and prepare a research proposal. They will present the proposal in the class. The teacher will provide feedback on their works.</li> </ul>                                                                        |

## 5. Evaluation

### 5.1 Internal Evaluation 40%

Internal evaluation will be conducted by subject teacher based on following activities:

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|--------------------------------------------------------------|----|
| • Attendance                                                 | 5  |
| • Class participation                                        | 5  |
| • First assignment (Individual work based on Unit III)       | 10 |
| • Second assignment (Group work based on Unit II, IV and V)  | 10 |
| • Third assignment (Written test: objectives and subjective) | 10 |
| • Total                                                      | 40 |

### 5.2 Final/Semester Evaluation 60%

Examination Division, Office of the Dean, Faculty of Education will conduct final examination at the end of semester. The number of questions and allocated marks are given below:

|                                                        |      |
|--------------------------------------------------------|------|
| • Objective type question (multiple choice 10 x 1mark) | = 10 |
| • Short answer questions (6 questions x 5 marks)       | =30  |
| • Long answer questions (2 questions x 10 points)      | =20  |
| 20                                                     |      |
| Total                                                  | 60   |

### Recommended books

Ary, D., Jacobs, L.C., Sorensen, C. & Walker, D.A. (2014). *Introduction to research in education* (9<sup>th</sup> ed.). Belmont, CA: Wadsworth (Unit 1 to 4).

Gay, L.R., Mills, G.E. & Airasian, P. (2006). *Educational research: competencies for analysis and applications* (8<sup>th</sup> ed.). Upper Saddle River, NJ: Pearson (Unit 1 to 5).

Best, J.W. and Kahn, J.V. (2006). *Research in education* (10<sup>th</sup> ed.). Upper Saddle River, NJ: Pearson/Allyn and Bacon (Unit 1 to 5)

### **Reference books**

Altrichter, H., Posch, P. &Somekh, B. (1993). *Teachers investigate their work: An introduction to the methods of action research*. London: Routledge

Creswell, J. W. (2012). *Educational research: Planning, conducting and evaluating quantitative and qualitative research* (4<sup>th</sup> ed.). Boston: Pearson.